Gene expression kinetics of *Exaiptasia pallida* innate immune response to *Vibrio parahaemolyticus* infection

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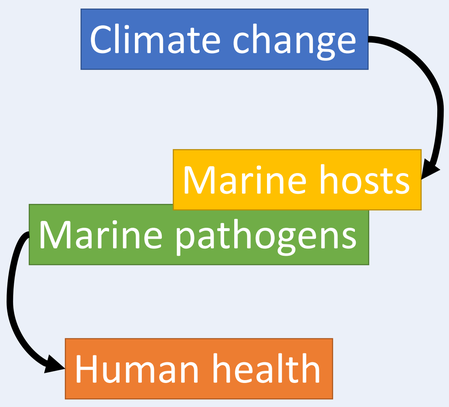
Introduction

Climate change acts on the physical conditions of the ocean. These environmental changes have the greatest of influence on micro-organisms including **marine bacteria** at the basis of all ecosystems, and therefore the food-chains that feed us. With warmer sea surface temperatures and coastal pollutions, the risks for marine bacteria to become pathogenic for human also increases. In order to better study these relationships, we are developing a marine host/pathogen model system using the sea anemone ***Exaiptasia pallida*** (Ep) as host and the ***Vibrio parahaemolyticus*** (**Vp**) clinical strain 03:K6 as marine bacteria pathogenic for humans. Ep's genome is rich with **innate immunity gene families** that have evolved over ~600 Myrs and show significant homologies with human counterparts, making the anemone a valuable animal model to study innate immunity. Vp occurs naturally within the bacterial community of cnidarians, but is also the first cause of food-borne gastroenteritis world-wide via the consumption of contaminated seafood. Therefore, the same Vp strain can either be used in seawater or human cell culture conditions, making it a powerful infectious pathogen model.

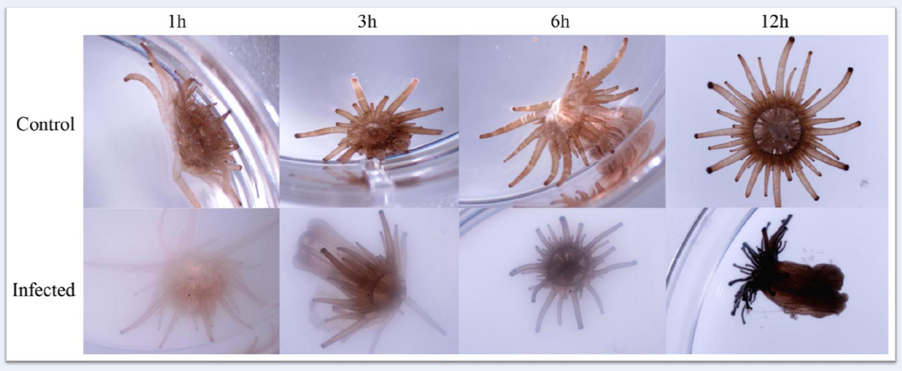


Methods

Anemone clones CC7 placed in 6 wells plates with 10 mL seawater were exposed to Vp at 10^8 cfu/mL for 1, 3, 6 and 12 hours at 27 °C. Transcriptomes of infected anemones were sequenced using RNA-seq and compared to controls. We performed pair-wise, time series and functional analyses to detect significant gene regulations and the molecular pathways significantly represented in the response of the anemone during the onset of infection.

Results

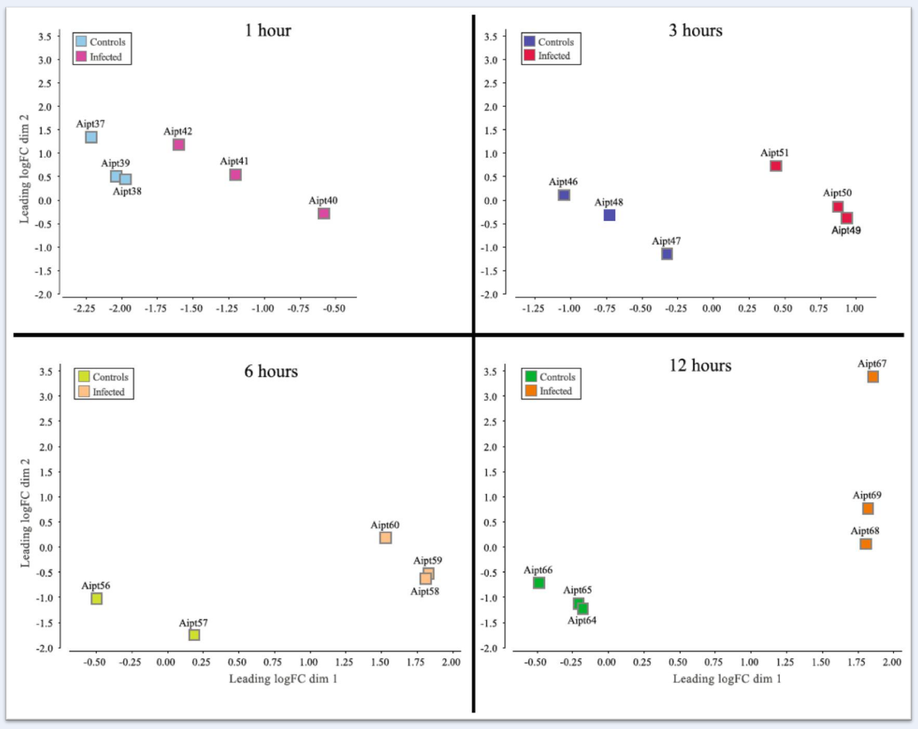
A Progression of Ep's infection with Vp



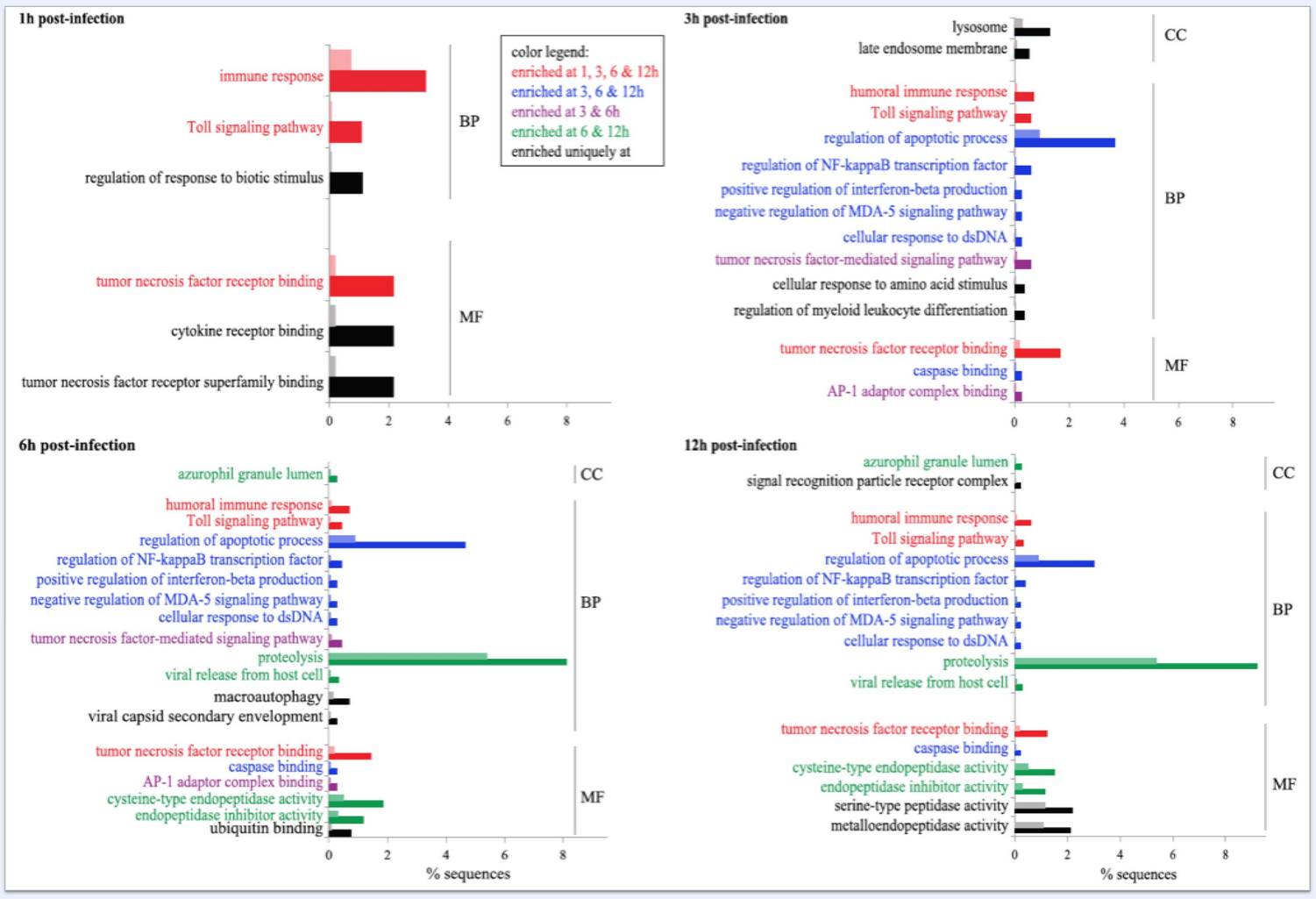
Ep's physiological response to Vp is observable over 12 hours in seawater (A) :
In 27 °C seawater, clinical strain Vp O3:K6 is still capable of infecting Ep. Infected anemones show mucus production, cell detachment, and shrinking. After 24 hours, Ep starts to die. However, if the infectious medium is removed after 3 or even 6 hours, anemones are capable of recovering over 48 hours. **Therefore, the anemone's defenses are efficient !**

Ep's transcriptomic activity rapidly increases during the onset of infection (B) :
Differences in gene expression between infected and control anemones are clearly illustrated by the separation of markers on the x-axis. The effect of time can also be seen on the y-axis. **Most of the gene regulation occurs over 6 hours.**

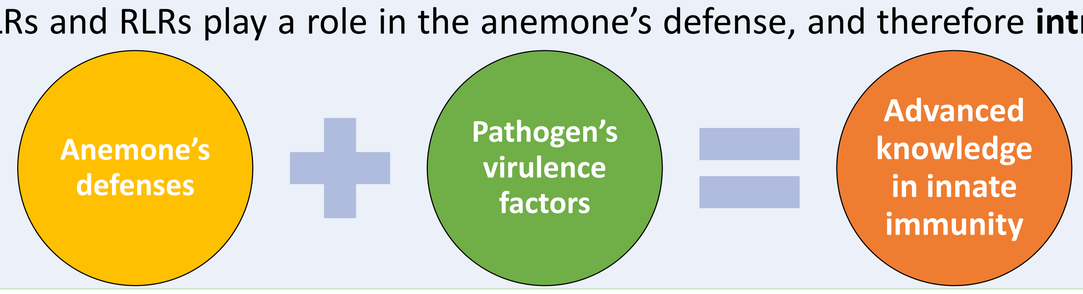
B Ep's transcriptomic response over 12 hours



C Innate immunity enrichment within up-regulated genes



Strong support for innate immunity (C) :
Significant proportions of genes linked to TNFRs, TLR-to-NF-κB, NLRs, RLRs, and apoptosis pathways are detected among positively regulated genes.
Functional conservation of human innate immune homologs is supported at the transcript level.

Conclusions

This study highlights, for the first time, the innate immunity gene candidates that respond to bacterial infection in a cnidarian model. Some of the hallmarks of innate immunity in humans were significantly regulated in Ep, and partially match the TLR-to-NF-κB pathway. Less studied pattern-recognition receptor families such as the NLRs and RLRs play a role in the anemone's defense, and therefore **introduce Ep as a valuable animal model to further study innate immunity.**